Exploring individual and community-level variation in quantifier scales

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1. Introduction : Quantifier Scales

1. Set Theoretic Definitions: (Barwise and Cooper, 1981)

⟦most⟧ $= \lambda P_{<e,t>}.\lambda Q_{<e,t>}.$ |P ∩ Q|>|P−Q| (more than half)

⟦many⟧ $= \lambda P_{<e,t>}.\lambda Q_{<e,t>}.|P \cap Q| > n$ (more than n)

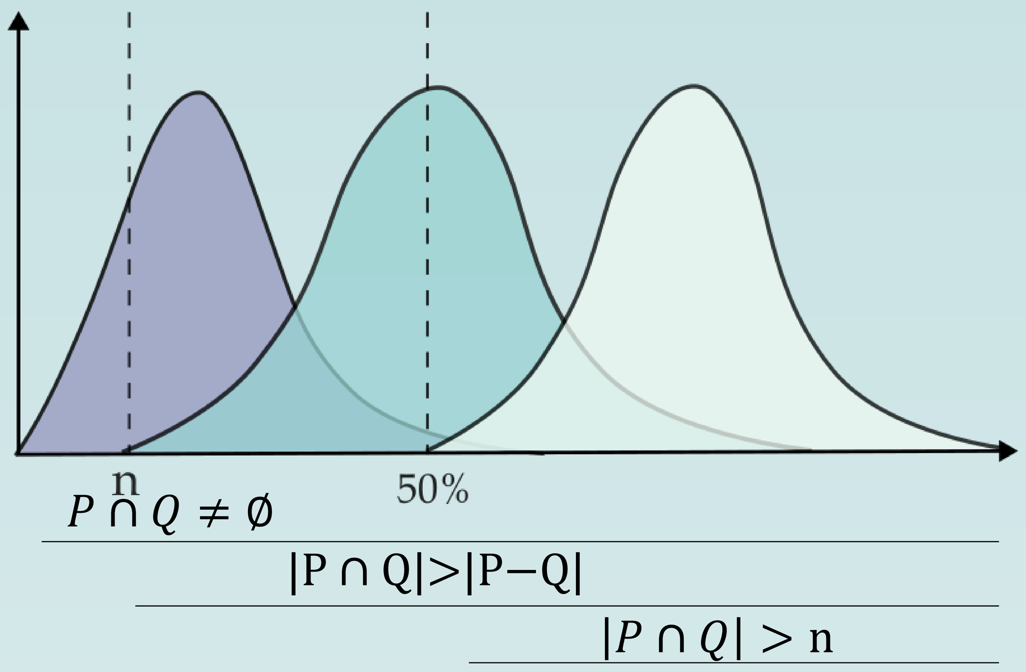
⟦some⟧ $= \lambda P_{<e,t>}.\lambda Q_{<e,t>}.P \cap Q \neq \emptyset$ (at least one)

2. Entailments: most ⇒ many ⇒ some; most ⇍ many ⇍ some;

3. Lexical Scales: <some, many, most>

4. Scalar Implicature (SI): *not many* (Horn 1972)

Quantifier usage is more restricted than their set theoretic definition



2. Research Questions

1. How does the quantifier *a few* enter into this picture?

Different analyses for *a few* in the literature (Solt 2006):

a. ⟦a few⟧ $= \lambda P_{<e,t>}.\lambda Q_{<e,t>}.P \cap Q \neq \emptyset$ Upper boundary Scalar Implicature

b. ⟦a few⟧ $= \lambda P_{<e,t>}.\lambda Q_{<e,t>}.|P \cap Q| < n$ Upper boundary lexically encoded

Entailments: a few ⇒ *some*;

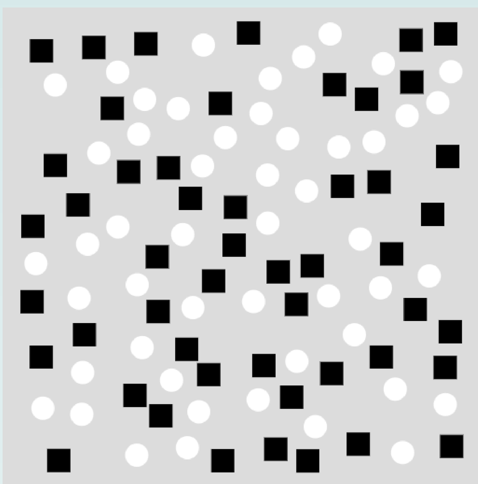
a. some ⇒ a few Should not form a scale

b. some ⇏ a few Should form a scale: <a few, some>

2. How does the space of quantifier alternatives vary across different Romance languages and speakers?

- Individual level variation in SI (Khorsheed and Gotzner 2023)
- Differences across quantifiers in terms of SI (McNabb 2015)
- Variation in children (Pagliarini et. al 2018)
- Quantifiers' alternatives vary diachronically (Bech 2024)

3. Methodology



Choose the option that best describes the image:
[**Most/Many/Some/A few**] objects are squares.
Approximate proportion of black squares: [**11%-20%, 21%-30%**...]

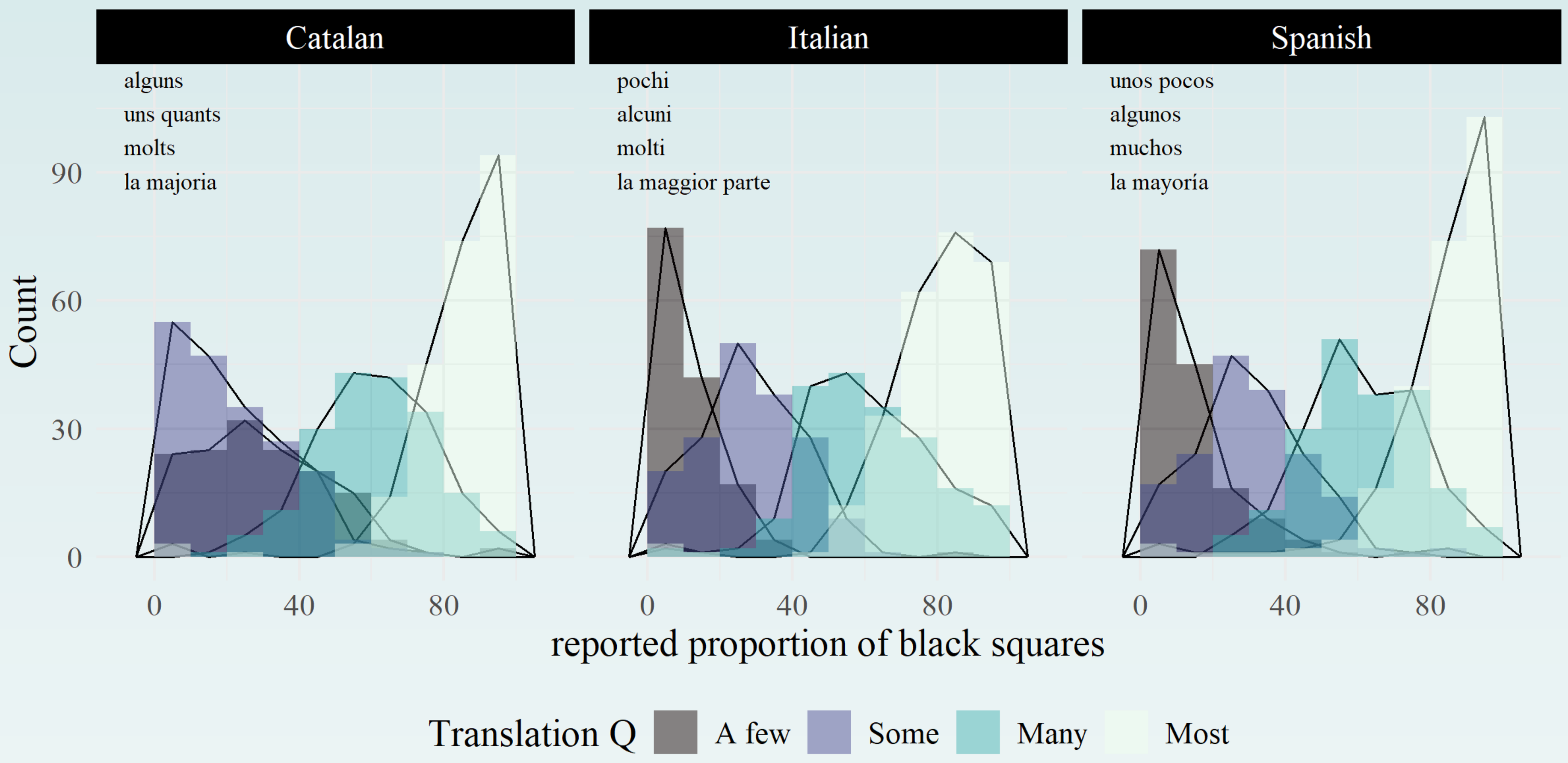
Spanish: *unos pocos, algunos, muchos, la mayoría*
Catalan: *uns quants*, alguns, molts, la majoria*
Italian: *pochi, alcuni, molti, la maggior parte*

*In Catalan *uns pocs* shows variability in the acceptability judgements

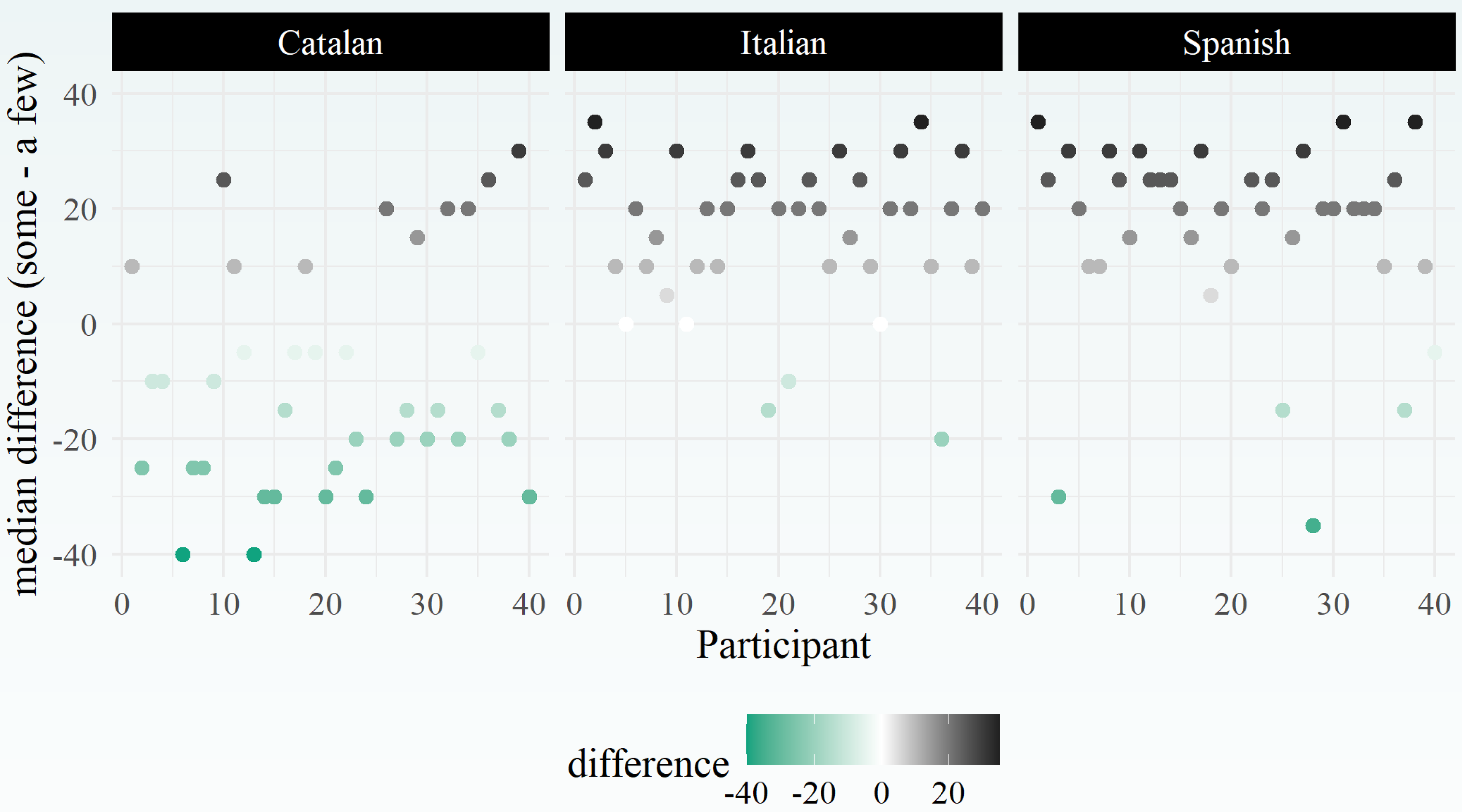
40 native spakers from each language recruited via Prolific. The study was conducted online in PCIbex Farm (Zehr & Schwarz 2018)

4. Results

Variation at the population level:



Variation at the individual level:



5. Discussion

Variation in Quantifiers at the population level:

- In **Catalan** *uns quants* and *some* used in similar contexts

→ **a.** ⟦uns quants⟧=⟦alguns⟧; <uns quants/alguns, molts>

- In **Spanish** and **Italian** *unos pocos* and *pochi* are used to represent a smaller magnitude than *algunos* and *alcuni*

→ **b.** ⟦unos pocos⟧ ≠ ⟦algunos⟧; <unos pocos, algunos, muchos>

→ **b.** ⟦pochi⟧ ≠ ⟦alcuni⟧; <pochi, alcuni, molti>

Variation at the individual level:

There is considerable amount of intra-speaker variation (especially in the case of Catalan) and this could indicate that we are dealing with a case of **covert representational variation** (MacKenzie 2019).

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